

IMPORTANCE OF SENSORS In The IoT

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Different types of applications require different types of sensors to collect data from the environment. This article takes a look at some common IoT sensors

In an Internet of Things (IoT) ecosystem, two things are very important: the Internet and physical devices like sensors and actuators. As shown in Fig. 1, the bottom layer of the IoT system consists of sensor connectivity and network to collect information. This layer is an essential part of the IoT system and has network connectivity to the next layer, which is the gateway and network layer.

The main purpose of sensors is to collect data from the surrounding environment. Sensors, or 'things' of the IoT system, form the front end. These are connected directly or indirectly to IoT networks after signal conversion and processing. But all sensors are not the same and different IoT applications require different types of sensors. For instance, digital sensors are straightforward and easy to interface with a microcontroller using Serial Peripheral Interface (SPI) bus. But for analogue sensors, either analogue-to-digital converter (ADC) or Sigma-Delta modulator is used to convert the data into SPI output.

Some common types of IoT sensors

Temperature sensors. These devices measure the amount of heat energy generated from an object or surrounding area. They find application in air-conditioners, refrigerators and similar devices used for envi-

ronmental control. They are also used in manufacturing processes, agriculture and health industry.

Temperature sensors can be used almost in every IoT environment, from manufacturing to agriculture. In manufacturing, sensors are used to monitor the temperature of machines. In agriculture, these can be used to monitor the temperature of soil, water and plants.

Temperature sensors include thermocouples, thermistors, resistor temperature detectors (RTDs) and integrated circuits (ICs). Some common types of temperature sensors are shown in Fig. 2.

Humidity sensors. The amount of water vapour in air, or humidity, can affect human comfort as well as many manufacturing processes in industries. So monitoring humidity level is important. Most commonly used units for humidity measurement are relative humidity (RH), dew/frost point (D/F PT) and parts per million (PPM).

Motion sensors. Motion sensors are not only used for security purposes but also in automatic door controls, automatic parking systems, automated sinks, automated toilet flushers, hand dryers, energy management systems, etc. You use these sensors in the IoT and monitor them from your smartphone or computer. HC-SR501 passive infrared (PIR) sensor is a popular motion sensor for hobby projects.

Gas sensors. These sensors are used to detect toxic gases. The sensing technologies most commonly used are electrochemical, photo-ionisation and semiconductor. With technical advancements and new specifications, there are a multitude of gas sensors available to help extend the wired and wireless connectivity deployed in IoT applications.

Smoke sensors. Smoke detectors have been in use in homes and industries for quite a long time. With the advent of the IoT, their application has become more convenient and user-friendly. Fur-

Fig. 1: IoT architecture layers (Courtesy: www.coursera.org)

